

Lec 12

Unsupervised Learning

Networks

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- unsupervised learning. In this learning, there exists no feedback from the system (environment) to indicate the desired outputs of a network. The network by itself should discover any relationships of interest, such as features, patterns, contours, correlations or categories, classifications in the input data, and thereby translate the discovered relationships into outputs

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- The process for achieving this is called competition. Practically, considering a set of students, if we want to classify them on the basis of evaluation performance, their score may be calculated, and the one whose score is higher than the others should be the winner.
- The extreme form of these competitive nets is called winner-take-all. The name itself implies that only one neuron in the competing group will possess a nonzero output signal at the end of competition.

- There exist two methods to determine the winner of the network during competition.
- **One** of the methods for determining the winner uses the square of the Euclidean distance between the input vector and weight vector, and the unit whose weight vector is at the smallest Euclidean distance from the input vector is chosen as the winner.
- **The next method** uses the dot product of the input vector and weight vector. The dot product between the input vector and weight vector is nothing but the net inputs calculated for the corresponding cluster units. The unit with the largest dot product is chosen as the winner and the weight updation is performed over it because the one with largest dot product corresponds to the smallest angle between the input and weight vectors, if both are of unit length.

FIXED WEIGHT COMPETITIVE NETS

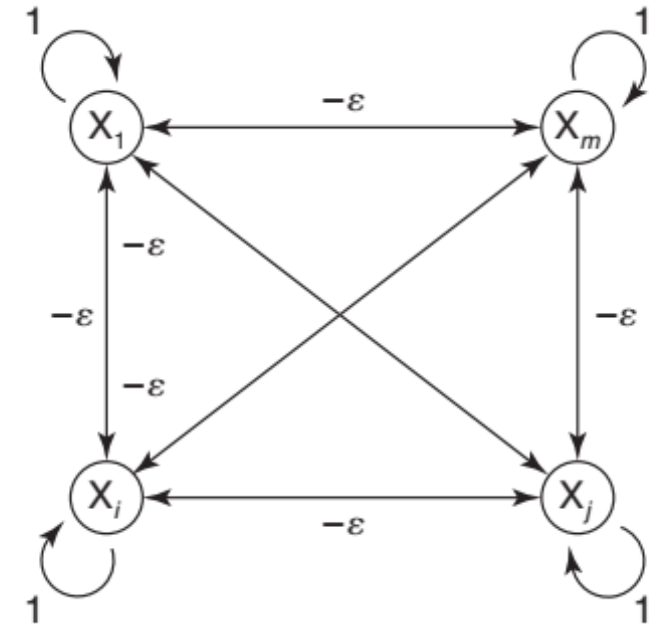
- **Maxnet**

- All the nodes present in this subnet are fully interconnected and
- there exist symmetrical weights in all these weighted interconnections. As such, there is no specific algorithm to train Maxnet; the weights are fixed in this case.

fixed symmetrical weights are present over the weighted interconnections.

The weights between the neurons are inhibitory and fixed.

The Maxnet with this structure can be used as a subnet to select a particular node whose net input is the largest.



Step 0: Initial weights and initial activations are set. The weight is set as $[0 < \varepsilon < 1/m]$, where “ m ” is the total number of nodes.

Let

$$x_j(0) = \text{input to the node } X_j$$

and

$$w_{ij} = \begin{cases} 1 & \text{if } i = j \\ -\varepsilon & \text{if } i \neq j \end{cases}$$

Step 1: Perform Steps 2–4, when stopping condition is false.

Step 2: Update the activations of each node. For $j = 1$ to m ,

$$x_j(\text{new}) = f \left[x_j(\text{old}) - \varepsilon \sum_{i \neq j} x_k(\text{old}) \right]$$

The Maxnet uses the following activation function:

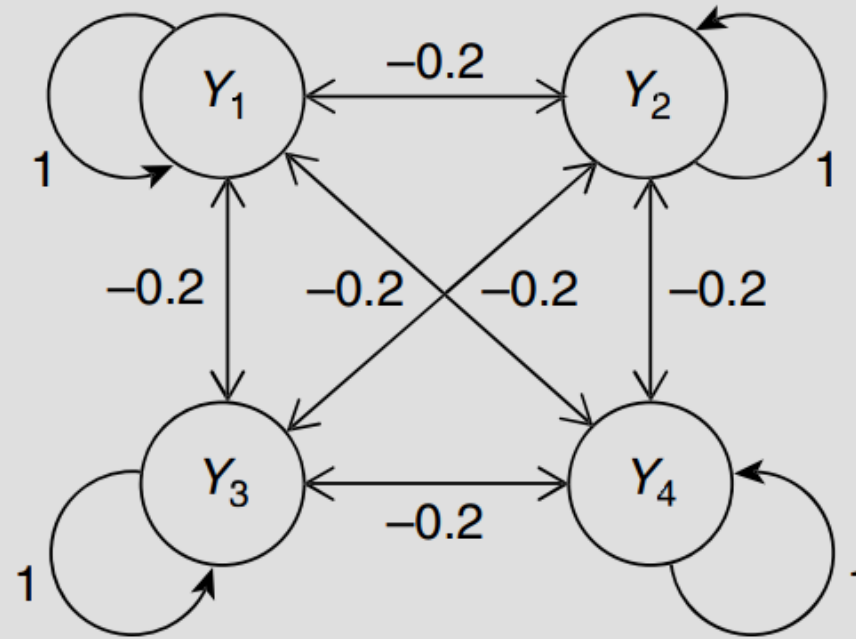
$$f(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

Step 3: Save the activations obtained for use in the next iteration. For $j = 1$ to m ,

$$x_j(\text{old}) = x_j(\text{new})$$

Step 4: Finally, test the stopping condition for convergence of the network. The following is the stopping condition: If more than one node has a nonzero activation, continue; else stop.

Let us consider the 4-unit MAXNET shown in Fig. 9.4. The inhibitory weight is taken as $\delta = 0.2$ which satisfies the condition $0 < \delta < \frac{1}{4}$. The input pattern $\mathbf{x} = [x_1, x_2, x_3, x_4] = [0.5, 0.8, 0.3, 0.6]$ is to be clustered. The step-by-step execution of **Algorithm** MAXNET-Clustering is given below.



Kohonen self-organizing feature map

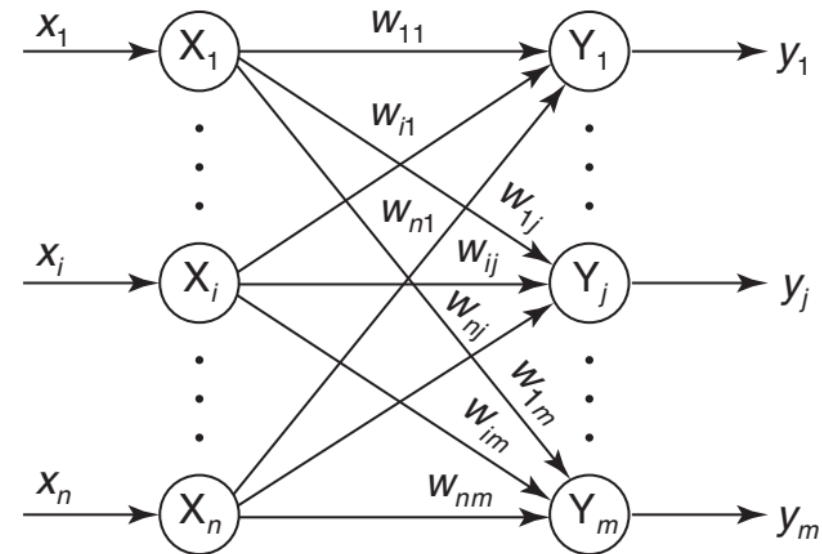
- Feature mapping is a process which converts the patterns of arbitrary dimensionality into a response of one- or two dimensional arrays of neurons, i.e., it converts a wide pattern space into a typical feature space.
- The network performing such a mapping is called feature map. Apart from its capability to reduce the higher dimensionality, it has to preserve the neighbourhood relations of the input patterns, i.e., it has to obtain a topology preserving map. For obtaining such feature maps, it is required to find a self-organizing neural array which consists of neurons arranged in a one-dimensional array or a two-dimensional array.

Kohonen self-organizing feature map

- A typical architecture of Kohonen self-organizing feature map (KSOFM) is shown in Figure

The architecture consists of two layers: input layer and output layer (cluster). There are “ n ” units in the input layer and “ m ” units in the output layer.

Basically, here the winner unit is identified by using either dot product or **Euclidean distance** method and the **weight updation using Kohonen learning rules** is performed over the winning cluster unit.



Training Algorithm

- Step 0:**
- Initialize the weights w_{ij} : Random values may be assumed. They can be chosen as the same range of values as the components of the input vector. If information related to distribution of clusters is known, the initial weights can be taken to reflect that prior knowledge.
 - Set topological neighborhood parameters: As clustering progresses, the radius of the neighborhood decreases.
 - Initialize the learning rate α : It should be a slowly decreasing function of time.

Step 1: Perform Steps 2–8 when stopping condition is false.

Step 2: Perform Steps 3–5 for each input vector x .

Step 3: Compute the square of the Euclidean distance, i.e., for each $j = 1$ to m ,

$$D(j) = \sum_{i=1}^n \sum_{j=1}^m (x_i - w_{ij})^2$$

Step 4: Find the winning unit index J , so that $D(J)$ is minimum. (In Steps 3 and 4, dot product method can also be used to find the winner, which is basically the calculation of net input, and the winner will be the one with the largest dot product.)

Step 5: For all units j within a specific neighborhood of J and for all i , calculate the new weights:

$$w_{ij}(\text{new}) = w_{ij}(\text{old}) + \alpha [x_i - w_{ij}(\text{old})]$$

or

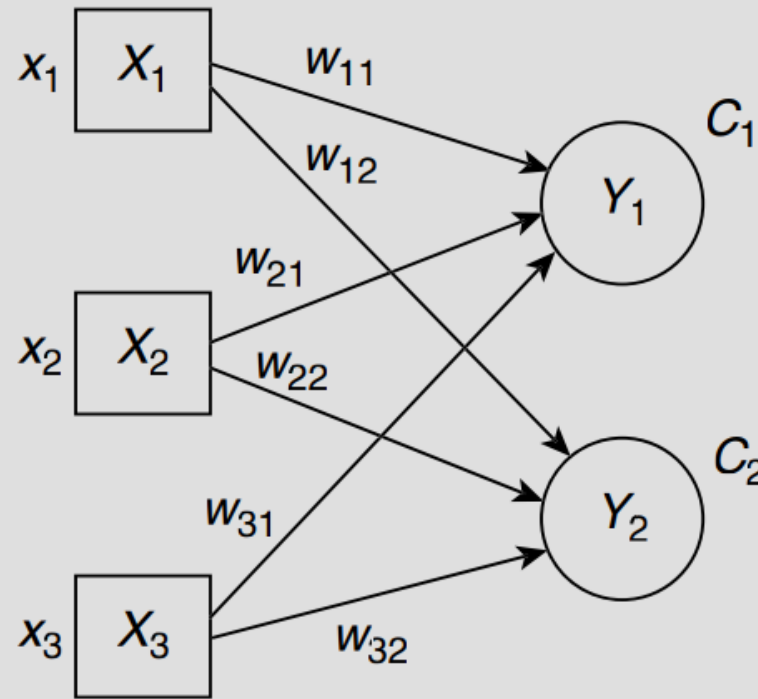
$$w_{ij}(\text{new}) = (1 - \alpha)w_{ij}(\text{old}) + \alpha x_i$$

Step 6: Update the learning rate α using the formula $\alpha(t+1) = 0.5\alpha(t)$.

Step 7: Reduce radius of topological neighborhood at specified time intervals.

Step 8: Test for stopping condition of the network.

Suppose there are four patterns $s_1 = [1, 0, 0]$, $s_2 = [0, 0, 1]$, $s_3 = [1, 1, 0]$ and $s_4 = [0, 1, 1]$ to be clustered into two clusters. The target SOM, as shown in Fig. 9.9, consists of three input units and two output units.



$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \end{bmatrix}$$

$$W = \begin{bmatrix} .5 & .3 \\ .8 & .5 \\ .4 & .3 \end{bmatrix}$$